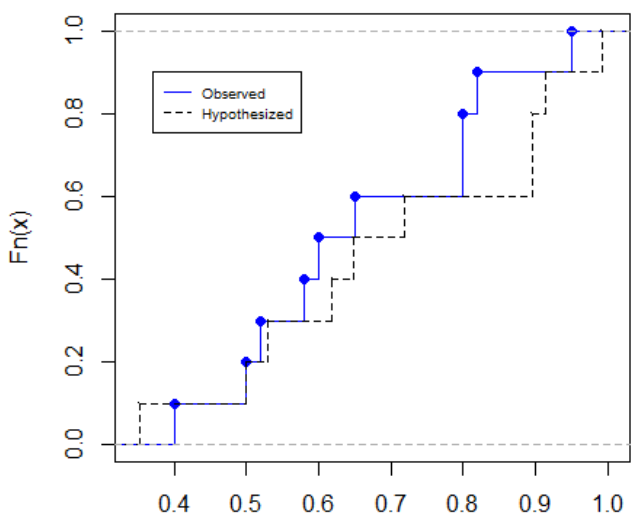
Adrian Cuyugan **2014-91450** **Stat 266** **Test of Goodness of Fit: Komolgorov-Smirnov One Sample Statistic**

Problem 4.34 - Ten students take a test and their scores (out of 100) are as follows:
95, 80, 40, 52, 60, 80, 58, 65, 50, 80

Test the null hypothesis that the cumulative distribution function of the proportion of right answers a student gets on the test is:

$$F_0(x) = \begin{cases} 0 & x < 0 \\ x^2(3-2x) & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

i	x (ordered)	$S_n(x)$	$F_0(x)$	$S_n(x) - F_0(x)$	$S_n(x-\varepsilon) - F_0(x)$	$\lvert S_n(x) - F_0(x)\rvert$	$\lvert S_n(x-\varepsilon) - F_0(x)\rvert$	$S_n(x) - D_{n,\alpha}$	$S_n(x) + D_{n,\alpha}$
1	0.4000	0.1000	0.3520	-0.2520	-0.3520	0.2520	0.3520	0.0000	**0.5090**
2	0.5000	0.2000	0.5000	-0.3000	-0.4000	0.3000	**0.4000**	0.0000	0.6090
3	0.5200	0.3000	0.5300	-0.2300	-0.3300	0.2300	0.3300	0.0000	0.7090
4	0.5800	0.4000	0.6190	-0.2190	-0.3190	0.2190	0.3190	0.0000	0.8090
5	0.6000	0.5000	0.6480	-0.1480	-0.2480	0.1480	0.2480	0.0910	0.9090
6	0.6500	0.6000	0.7183	-0.1183	-0.2183	0.1183	0.2183	0.1910	1.0000
7	0.8000	0.8000	0.8960	-0.0960	-0.2960	0.0960	0.2960	0.3910	1.0000
8	0.8000	0.8000	0.8960	-0.0960	-0.0960	0.0960	0.0960	0.3910	1.0000
9	0.8200	0.9000	0.9145	-0.0145	-0.1145	0.0145	0.1145	0.4910	1.0000
10	0.9500	1.0000	0.9928	0.0072	-0.0928	0.0072	0.0928	**0.5910**	1.0000



$H_0: F_x(x) = F_0(x)$ for all x, $x^2(3-2x)$
$H_A: F_x(x) \neq F_0(x)$ for some x

To calculate the KS statistic, $D_n = \sup_x |S_n(x) - F_0(x)| = \max_x [|S_n(x) - F_0(x)|, |S_n(x-\varepsilon) - F_0(x)|]$ = 0.40

Significance level is set at $\alpha = 0.05$. In Table F, the rejection region of 0.05, $n = 10$ is $D_{n,\alpha} \geq 0.409$. Since the calculated KS statistic does not exceed of the tabular value, the null hypothesis is not rejected; following a distribution of $x^2(3-2x)$ from the proportion of correct answers a student gets on a test.

P-value method also does not reject the null hypothesis at 0.05 level of significance:
$0.05 < P[D_n \geq 0.40 \mid H_0] < 0.10$

The confidence bands are as follows:
$L_n(x) = \max[S_n(x) - D_{n,a}, 0] = 0.591$ and $U_n(x) = \min[S_n(x) + D_{n,a}, 1] = 0.509$

```r
ks.observed <- c(0.40, 0.50, 0.52, 0.58, 0.60, 0.65, 0.80, 0.80, 0.82, 0.95)
ks.hypothesized <- (ks.observed^2)*(3-2*ks.observed) #x^2(3-2x)
ks.df <- cbind.data.frame(ks.observed, ks.hypothesized) # Combines two vectors in a dataframe format
ks.df # Views the dataset
```

```
##    ks.observed ks.hypothesized
## 1         0.40        0.352000
## 2         0.50        0.500000
## 3         0.52        0.529984
## 4         0.58        0.618976
## 5         0.60        0.648000
## 6         0.65        0.718250
## 7         0.80        0.896000
## 8         0.80        0.896000
## 9         0.82        0.914464
## 10        0.95        0.992750
```

```r
# Plots the empirical cumulative distribution
plot(ecdf(ks.observed), pch = 19, verticals = T,col='blue', main = NULL, xlab = '')
lines(ecdf(ks.hypothesized), verticals = T, pch = 46, lty = 2)
legend(x = 0.37, y = 0.9, legend=c("Observed", "Hypothesized"),
       lty = c(1,2), col = c('blue', 'black'), cex = 0.6)
```

```r
# Runs KS test using two-sample but the other population is based on the hypothesized distribution
ks.test(x = ks.observed, y = ks.hypothesized)
```

```
## Warning in ks.test(x = ks.observed, y = ks.hypothesized): cannot compute
## exact p-value with ties
```

```
##
##  Two-sample Kolmogorov-Smirnov test
##
## data:  ks.observed and ks.hypothesized
## D = 0.3, p-value = 0.7591
## alternative hypothesis: two-sided
```